

program is loaded into the chip using PICKIT2 programmer. The file can be imported by the PICKIT2 programmer as a hex file and the configuration byte entered by clicking Configuration option.

H-Bridge circuit

Before you decide on the DSP embedded controller and the way it is connected and programmed to generate such waveforms of pulses, let us take a look at the way sine wave current pulses are sent to the motor and its windings. The winding must receive current pulses for both positive and negative half cycles. But we generate only two sets of pulses and these must be passed into the motor winding in the forward and reverse directions during the positive and negative half cycles, respectively. In other words, you have to reverse the direction of current for the negative half-cycle pulse train.

For this, the motor windings are connected in the form of an H-bridge (as shown in Fig. 4) to the DC supply with switches to direct the pulses. In Fig. 4, the mains voltage (170V to 240V) is rectified by two diodes and converted into positive and negative DC voltages with respect to the neutral line of the AC supply. The circuit is only a half-wave rectifier. During positive half cycles, diode D1 conducts and charges C1. During negative half cycles, diode D2 conducts to charge capacitor C2. Both capacitors are large-value electrolytic capacitors, each of which must be at least 470µF, 400V in order to avoid considerable ripple on the direct current rails.

Thus, you have two DC rails, one for positive and the other for negative voltages. When the mains supply (say, 200V) is rectified by a diode, the peak voltage of the capacitor is $200 \times \sqrt{2} = 282\text{V}$. So C1 and C2 are high-voltage (400V) electrolytic capacitors. All computer SMPS boards use these invariably.

The H-bridge configuration used

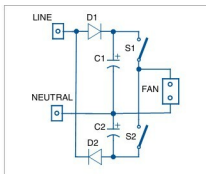


Fig. 4: Semi H-bridge circuit with the load (fan)

here is actually a semi-H-bridge. Like a Wheatstone bridge, it has four arms. The two arms on the left side have capacitors C1 and C2, with the centre junction taken out. The right side consists of two switches (S1 and S2) one below the other connected between the positive and negative DC rails. There is a junction point between the two switches. The load, the fan in our case, is to be connected between the two junctions.

Switches S1 and S2 in Fig. 4 are actually transistors (bipolar, MOSFET or IGBT). You can select suitable transistors according to the rating needed and at the lowest cost.

At a time, only one of the switches is turned on. Let's say, the waveform of pulse sequence shown in Fig. 1 is used to turn on switch S1. Current will flow from the positive side to the switch and then through the fan load to the junction point of C1 and C2. The flow of current is from right to left through the load. This current constitutes the positive half cycle.

When switch S2 is turned on using the second set of pulse sequence, the current will flow from left to right side of the load. This current constitutes the negative half cycle.

In this manner, the alternating current of the frequency generated by the PWM waves will be applied to the fan. When you apply 60Hz, the motor speed increases. Using the microcontroller, you can vary the frequency from 40 to 70Hz. It is not

advisable to increase the frequency for the fan motor beyond this value.

Generating the AC sine voltage at variable frequency

The variable-frequency sinusoidal waveform is generated for positive and negative half cycles by closing switches S1 and S2 alternately. S1 is switched on and off as per the pulses from PWM1, whose width varies as per Fig. 1 to fit the sine wave shape. This goes for one half cycle time, which is 20ms for 50Hz current, 16ms for 60Hz current and so on. Then, the next half cycle is obtained by sending current, using switch S2, in the opposite direction through the motor coil. Switch S2 is also repeatedly closed and opened as per the pulses coming from PWM2 pin.

Now the main challenge is to choose the transistors for S1 and S2. If IGBTs or MOSFETs are used, switching S1 'on' is not easy. So bipolar transistor version is chosen here. You can preferably use Darling-ton bipolar transistor rated 350V or more. Transistor MJ5742 is a good choice with its voltage rating of 400V, gain of 100 and maximum current rating of 8A. But, it is expensive. So a compromise has to be made.

In order to switch on the transistors using PWM pulses, use of optoisolators is necessary. Optoisolators have an LED and a photo-transistor in one package, usually 6-pin DIP. When the pulses of current are sent to the LED in the optoisolator through a resistor, the phototransistor inside turns on. This transistor is connected across the collector to the base of the switching transistor. So, each pulse turns on the switch for the duration equal to the pulse width and passes current through the motor coil connected between the two junction points—switches and capacitors as in Fig. 4.

The choice of the optoisolator is critical in two ways. First, it should have a phototransistor resistance such that adequate current flows